

Reciprocating Air Compressors

16.1 Introduction

Air compressors are used for increasing pressure of air by compression. The compressed air has many practical uses such as (i) driving a compressed air engine; (ii) driving pneumatic tools; (iii) paint spraying; (iv) injecting fuel in Diesel engines; (v) for boosting internal combustion engines; (vi) conveying solid and powdered materials in pipe lines; (vii) transmitting control-system pressures to remote locations; (viii) in blast furnaces and boiler furnaces; (ix) cleaning surfaces by air blast; (x) cooling of large buildings, etc. In mixing industries it is used where fire risks may preclude the use of internal combustion engines or electric motors.

16.2 Classification

The compressors may be classified as follows :

(a) **Design and principle of operation.** There are two basic types of air compressors.

(i) Positive displacement compressors, and (ii) Non-positive or steady flow compressors.

(i) **Positive displacement compressors.** These are further classified as reciprocating compressors and rotary compressors. In positive displacement compressors the fluid is prevented by a solid boundary from flowing back in the direction of the pressure gradient. These compressors are capable of producing large pressure ratios. Reciprocating machines are regarded as open systems, being steadily supplied with the fluid air. The work and heat transfer between the fluid and its surroundings is at a uniform rate. However the internal processes are intermittent. This means that these are best suited for comparatively small mass flows. In reciprocating machines the displacement is performed by a piston-cylinder as in steam or internal combustion engines. Like steam engines reciprocating compressors may be single-acting or double-acting.

In positive displacement rotary compressors the displacement is performed by a rotating component. The rotary machine can operate at high speeds and can therefore handle large mass flow than its reciprocating counterpart. There are many types of positive displacement rotary compressors, the chief among them being *Roots blower* and *Vane type blower*. For details of rotary positive displacement compressors see chapter on Rotary compressors.

(ii) **Non-positive compressors.** These are the rotary compressors of *centrifugal* and *axial flow design*. In these compressors the fluid is not contained by solid boundaries but is continuously in a steady flow through the machine, undergoing changes in pressure primarily by means of dynamic effects. The pressure range of atmospheric to about 9 bar is common to both types—positive and non-positive compressors.

(b) **Number of stages.** The compressor may be a single stage or multi-stage, and have either air or water cooling. Multi stage compression is done to achieve high pressure ratios. In a compressor when compression ratio exceeds 5, generally multi-stage compression is adopted. The following arrangements are generally in practice for reciprocating compressors :

- Single-stage compressors : delivery pressure upto 5 bar
- Two-stage compressors : delivery pressure 5 to 35 bar
- Three-stage compressors : delivery pressure 35 to 85 bar
- Four-stage compressors : delivery pressure above 85 bar

(c) **Pressure limits.** Compressors are also classified as per the delivery pressure.

- Low pressure compressors : delivery pressure upto 1 bar
 - Medium pressure compressors : delivery pressure 1 to 8 bar
 - High pressure compressors : delivery pressure 8 to 10 bar
 - Super-high pressure compressors : delivery pressure above 10 bar
- According to pressure rise limits the compressors are also classified as
- Fans : Pressure ratio is 1 to 1.1
 - Blowers : Pressure ratio is 1.1 to 2.5
 - Compressors : Pressure ratio is above 2.5

Compressors used for creating vacuum are known as *air pumps* or *exhausters*.

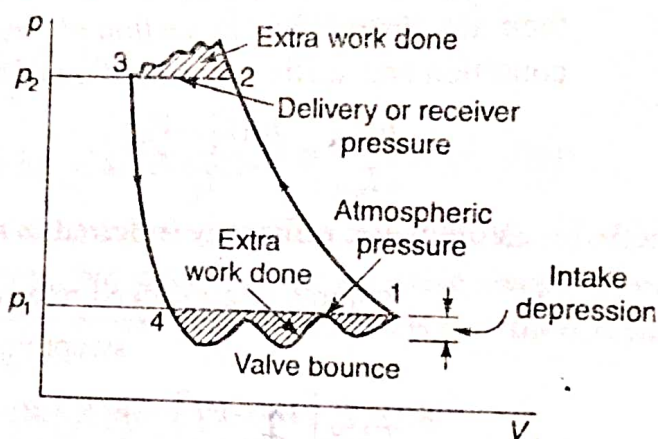
(d) **Capacity.** Compressors are also classified according to the volume of air delivered, per unit time.

- Low capacity compressors : volume delivering 0.15 m³/s or less
- Medium capacity compressors : volume delivered 0.15 m³/s to 5 m³/s
- High capacity compressors : volume delivered is above 5 m³/s

16.7 Actual indicator diagram

Actual indicator diagrams differ from hypothetical diagrams due to turbulence and eddies in air and inertia of automatic spring loaded suction and delivery valves as shown in Fig. 16.4. The inlet valve should theoretically open at 4, but actually it does not open because of the valve inertia and because there must be a pressure difference across the valve in order to move it. Thus the pressure drops off below atmospheric until the valve moves off its seat. The opening of the valve is followed by some valve bounce and eventually the intake will become steady at some pressure below atmospheric. This pressure difference is called *intake depression*. Similar situation occurs on the delivery side. The extra work done due to this may be called the *pumping work*. The area of the indicator diagram is the *indicated work* required. The actual power input is larger than the indicated power due to work necessary to overcome the losses due to friction, etc. The actual power input is called the *brake power* or *shaft power*.

Fig. 16.4 Actual indicator diagram



Mechanical efficiency of a reciprocating compressor is defined as the ratio of indicated power to the brake power of the shaft supplying power to the compressor.

$$\text{Mechanical efficiency} = \frac{\text{indicated air power}}{\text{brake (or shaft) power}} \quad (16.11)$$

The indicated power or air power is determined from the area of indicator diagram. In practice the average height is found from the indicator diagram and then knowing the pressure scale mean effective pressure is calculated.

$$\text{Mean effective pressure, } p_m = \frac{\text{area of diagram}}{\text{length of diagram}} \times \text{pressure scale} \quad (16.12)$$

and the indicated power, $i_p = p_m l A N$, in kW

where p_m is taken in kN/m^2 , A in m^2 , l in m and N in rev/s for a single-stage single acting compressor.

In many compressors the *overall isothermal efficiency* is defined by

$$\frac{\text{Isothermal power}}{\text{Brake power applied to the compressor}} \quad (16.13)$$

The value of overall isothermal efficiency is nearly 60%.

Gas Compressors

Learning Objectives

After reading this chapter you should be able to

- LO 18.1 Define the function of gas compressor and understand its types and applications
- LO 18.2 Explain adiabatic, polytropic and isothermal work on a reversible work of compression
- LO 18.3 Describe the arrangement and work transfer in a single-stage reciprocating air compressor
- LO 18.4 Calculate the volumetric efficiency expression and know its various controlling parameters
- LO 18.5 Explain the details of using multi-stage compression and its preference over single stage
- LO 18.6 Discuss the pressure-volume diagram for an air motor
- LO 18.7 Classify the rotary compressor and discuss the comparison between their principles and operations

A gas compressor is a device in which work is done on the gas to raise its pressure, with an appreciable increase in its density. The compression of gases is an important process in many power plants, refrigeration plants and industrial plants. Industrial uses occur in connection with compressed air motors for tools, air brakes for vehicles, servo-mechanisms, metallurgical and chemical processes, conveying of materials through ducts, transporting of natural gas and production of bottled gases.

Compressors can be (a) positive displacement machines like reciprocating compressors, Root's blower and vane-sealed machines and (b) turbine type like centrifugal and axial flow compressors.

LO 18.1

Define the function of gas compressor and understand its types and applications

18.1 COMPRESSION PROCESSES

A gas compression process may be either adiabatic or involving heat transfer, depending upon the purpose for which the gas is compressed. If the compressed gas is to be used promptly in an engine or in a combustion process, adiabatic compression is desirable so as to obtain the maximum energy in the gas at the end of compression. In many applications, however, the gas is not used promptly but is stored in a tank or receiver for use later when needed. The gas in the tank transfers heat to the surroundings so that when finally used it is at room temperature. In such cases, the objective of the compression and storage process is simply to increase the pressure of the gas without change of temperature. It will be shown below that if the gas is cooled during compression, the work required will be less than for adiabatic compression. Another advantage of cooling is the reduction in volume of gas and less pipe friction losses. Since cooling during the compression process is not very effective, *after-coolers* are often used to cool the gas leaving the compressor.

18.2 WORK OF COMPRESSION

The steady flow energy equation for a compressor (Fig. 18.1) gives

$$h_1 + Q = h_2 + W_x \quad (18.1)$$

neglecting the changes of potential and kinetic energy. From the property relation

$$Tds = dh - vdp \quad (18.2)$$

For a reversible process,

$$Q = \Delta h - \int vdp \quad (18.3)$$

Let us consider two particular idealised cases, namely, reversible adiabatic and reversible isothermal, as well as a general case of a reversible polytropic process ($pv^n = \text{constant}$). The paths of such processes are plotted in Fig. 18.2 for an ideal gas compressed from state 1 to state 2.

As shown in Section 7.14 or from Eqs. (18.1 to 18.3) for adiabatic compression of unit mass,

$$W_x = - \int vdp \quad (18.4)$$

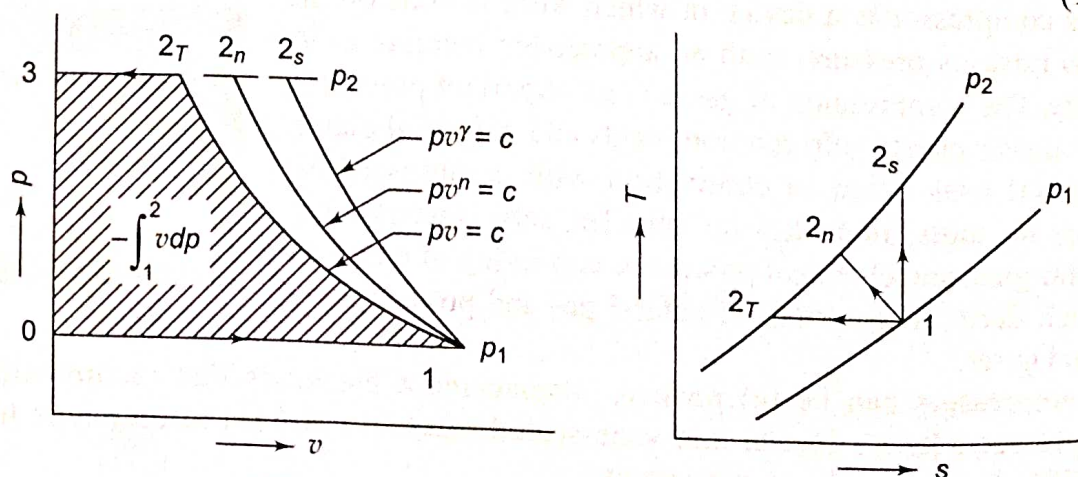


Fig. 18.2 Reversible compression processes

LO 18.2

Explain adiabatic, polytropic and isothermal work on a reversible work of compression

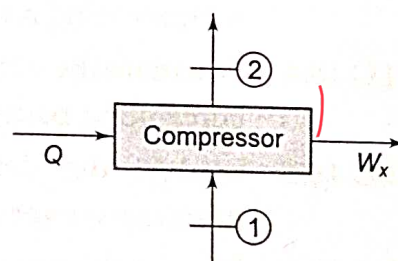


Fig. 18.1

For a reversible adiabatic process, $pv^\gamma = C$,

$$v = \left(\frac{pv^\gamma}{p} \right)^{\frac{1}{\gamma}} = \frac{p_1^{1/\gamma} v_1}{p^{1/\gamma}} = \frac{p_2^{1/\gamma} v_2}{p^{1/\gamma}}$$

and

$$\begin{aligned} W_x &= - \int_1^2 \frac{p_1^{1/\gamma} v_1}{p^{1/\gamma}} dp = - p_1^{1/\gamma} \cdot v_1 \cdot \frac{1}{1 - \frac{1}{\gamma}} \left[p_2^{1 - \frac{1}{\gamma}} - p_1^{1 - \frac{1}{\gamma}} \right] \\ &= - \frac{\gamma}{\gamma - 1} p_1 v_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{\gamma - 1}{\gamma}} - 1 \right] \end{aligned}$$

The work of compression or the steady flow work input to the gas is the negative of the shaft work W_x , i.e. for reversible adiabatic compression,

$$W_s = \frac{\gamma}{\gamma - 1} p_1 v_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{\gamma - 1}{\gamma}} - 1 \right] \quad (18.5)$$

Similarly, for reversible polytropic compression

$$W_n = \frac{n}{n - 1} p_1 v_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{n - 1}{n}} - 1 \right] \quad (18.6)$$

For reversible isothermal compression of an ideal gas,

$$W_t = - \int_1^2 v dp = - \int_1^2 \frac{p_1 v_1}{p} dp = p_1 v_1 \ln \frac{p_2}{p_1} \quad (18.7)$$

In the p - v plot of Fig. 18.2 the work of compression for each type of process is represented by the area between the path of that process and the axis of pressures.

For the general process $pv^n = C$, taking logarithm

$$\ln p + n \ln v = \ln C$$

On differentiation,

$$\frac{dp}{p} + n \frac{dv}{v} = 0$$

$\therefore$ The slope at any state (say 1) is given by

$$\frac{dp}{dv} = -n \frac{p_1}{v_1} \quad (18.8)$$

For $\gamma > n > 1$ and for the same pressure ratio p_2/p_1 , the isothermal compression needs the minimum work, whereas adiabatic compression needs the maximum work, while the polytropic compression needing work in between the two. In isothermal compression, all the work done on the gas during the process is transformed into an internal energy increase where $Q - W = u_2 - u_1 = 0$ (for an ideal gas) or $W = Q$.

$$\text{heat } \Delta T \cdot F = 0$$

This heat is taken away from the gas by cooling. Thus in isothermal compression, considered to be ideal, no energy would be imparted to the gas, since its function is simply to raise the pressure of the gas (and not its temperature).

The efficiency of a compressor working in a steady flow process may be defined as

$$\eta_c = \frac{h_{2s} - h_1}{W_c} = \frac{W_s}{W_c} \quad (18.9)$$

where W_c is the shaft work supplied to the compressor per unit mass. For the idealised reversible isothermal process the compressor efficiency is sometimes defined as

$$\eta_c = \frac{W_t}{W_c} \quad (18.10)$$

where W_t is the work of reversible isothermal compression. The two efficiencies of Eqs. (18.9) and (18.10) are called, respectively, the *adiabatic efficiency* and *isothermal efficiency*. Because of the effects of cooling, the adiabatic efficiency of a real compressor may be greater than unity.

Many turbine-type compressors are essentially adiabatic machines due to their high speeds. For an adiabatic machine the work of compression is equal to the enthalpy rise of the gas,

$$W_c = h_2 - h_1$$

Then for an adiabatic compressor the efficiency is

$$\eta_c = \frac{h_{2s} - h_1}{h_2 - h_1} \quad (18.11)$$

For a reciprocating machine the compressor efficiency may be on an indicated work basis or a brake work basis, depending upon where the work input is measured (See Section 3.3).

18.3 SINGLE-STAGE RECIPROCATING AIR COMPRESSOR

Figure 18.3 shows the arrangement of a single-stage reciprocating air compressor, together with a typical indicator diagram. The compressor operates on a two-stroke cycle as follows:

- Stroke 1 (a-c)** The piston withdraws, causing the air in the clearance volume to expand, and when the pressure in the cylinder falls below atmospheric pressure (at *b*) the inlet valve opens and air is drawn into the cylinder for the remainder of the stroke.
- Stroke 2 (c-a)** The piston moves inwards, compressing the air in the cylinder, and the inlet valve closes when the cylinder pressure reaches atmospheric pressure. Further compression follows as the piston moves towards the top of its stroke until, when the pressure in the cylinder is more than that in the receiver, the delivery valve opens and air is delivered to the receiver for the remainder of the stroke.

LO 18.3

Describe the arrangement and work transfer in a single-stage reciprocating air compressor

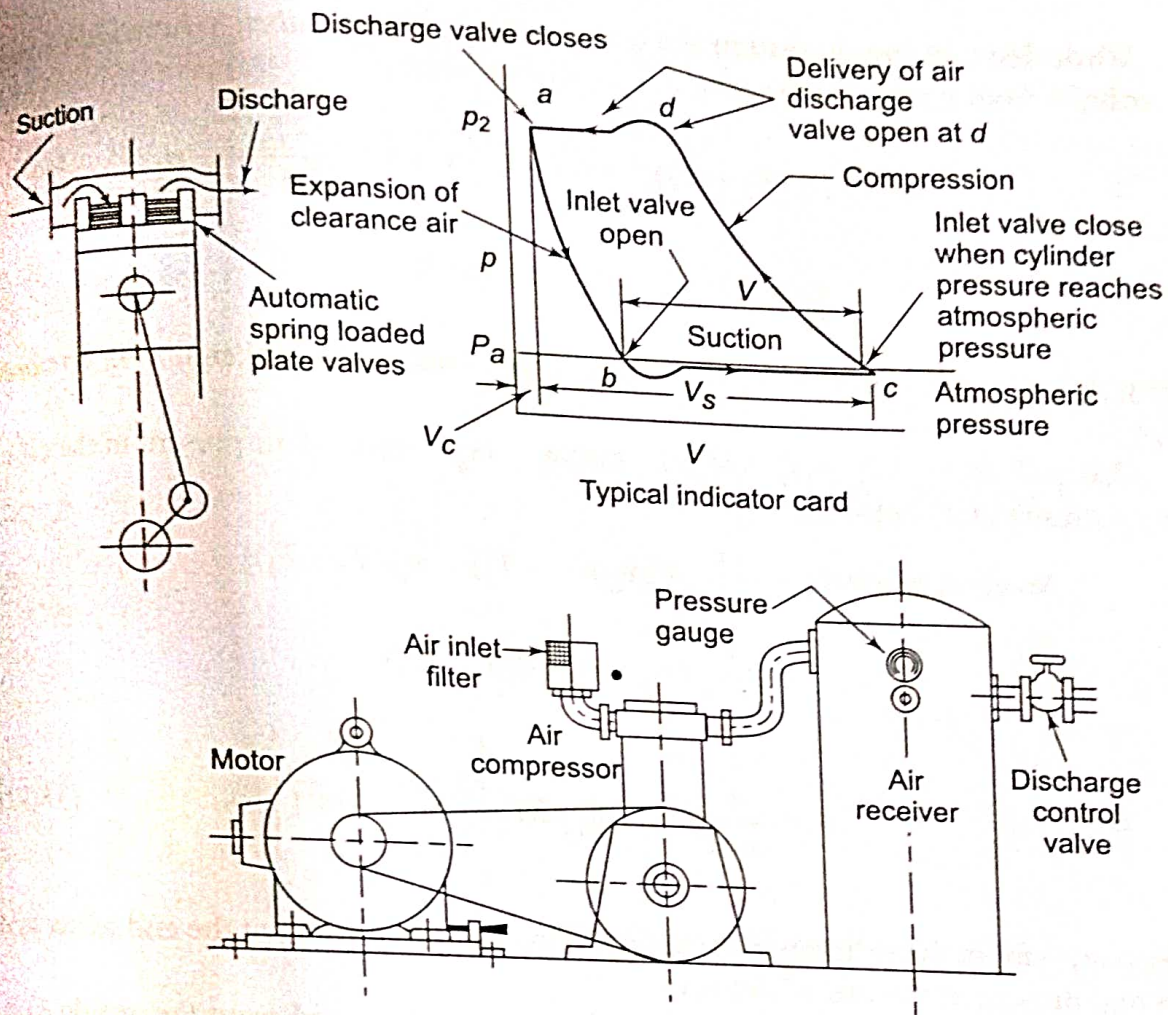


Fig. 18.3 Arrangement of single-stage reciprocating air compression

The intercept V on the indicator diagram represents the volume of air taken in per cycle. It is seen that the effect of clearance air is to reduce the quantity of air drawn in during the suction stroke, so that in practise the clearance space is made as small as possible.

The total area of the indicator diagram represents the actual work of the compressor on the gas or air. The areas above p_2 and below p_1 represent work done because of pressure drop through the valves and port passages; this work is called the valve loss. The idealised machine to which the actual machine is compared has an indicator diagram like Fig. 18.4. Both expansion and compression are supposed to follow the same law $pV^n = C$. The small quantity of high pressure air in the clearance volume expands to V_a and air drawn in during the suction stroke is $V_b - V_a$. From the diagram:

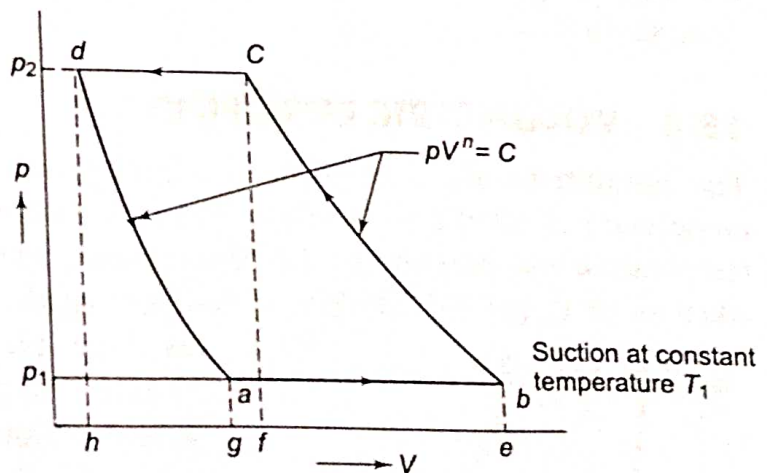


Fig. 18.4 Indicator diagram

Work done on the air during the cycle = Enclosed area $abcd$ = Area $dcfh$ + Area $cbef$ – Area $hgad$ – Area $geba$

$$\begin{aligned}\therefore W &= p_2 (V_c - V_d) + \frac{p_2 V_c - p_1 V_b}{n-1} - \frac{p_2 V_d - p_1 V_a}{n-1} - p_1 (V_b - V_a) \\ &= \frac{n}{n-1} [(p_2 V_c - p_1 V_b) - (p_2 V_d - p_1 V_a)]\end{aligned}$$

Now, $p_2 V_c = m_c R T_2$, $p_1 V_b = m_b R T_1$ and $m_c = m_b$ = mass of air taken during compression.

Also, $p_2 V_d = m_d R T_2$, $p_1 V_a = m_a R T_1$ and $m_a = m_d$ = mass of air present in the clearance volume after delivery.

$$\begin{aligned}\therefore \text{Work done/cycle} &= \frac{n}{n-1} R [m_b (T_2 - T_1) - m_d (T_2 - T_1)] \\ &= \frac{n}{n-1} (m_b - m_d) R (T_2 - T_1) \\ &= \frac{n}{n-1} (m_b - m_d) R T_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} - 1 \right] \quad (18.12)\end{aligned}$$

where $m_b - m_d$ is the difference between the mass of air present at the end of suction and that present at the end of delivery.

Thus the expression in Eq. (18.12) is the same as obtained from the steady flow energy equation given in Eq. (18.6) where the clearance volume was neglected. Thus the mass of gas in the clearance volume does not have any effect on the work of compression.

18.4 VOLUMETRIC EFFICIENCY

The amount of air dealt with in a given time by an air compressor is often referred to at free air conditions, i.e. the temperature and pressure of the environment, which may be taken as 15°C and 101.325 kPa, if not mentioned. It is known

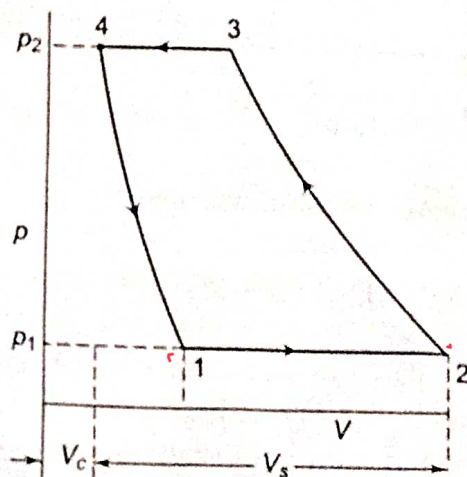


Fig. 18.5 Volumetric efficiency

as free air delivery (FAD). The ratio of the actual volume of gas taken into the cylinder during suction stroke to the piston displacement volume (PD) or the swept volume (V_s) of the piston is called the *volumetric efficiency*, or

$$\eta_{\text{vol}} = \frac{\dot{m} v_1}{PD} = \frac{\dot{m} v_1}{V_s}$$

where $\dot{m}$ is the mass flow rate of the gas and v_1 is the specific volume of the gas at inlet to the compressor. With reference to Fig. 18.5

LO 18.4

Calculate the volumetric efficiency expression and know its various controlling parameters

$$\eta_{\text{vol}} = \frac{V_2 - V_1}{V_s} = \frac{V_c + V_s - V_1}{V_s}$$

$$= 1 + \frac{V_c}{V_s} - \frac{V_1}{V_s}$$

Let $C = \text{clearance} = \frac{\text{Clearance volume}}{\text{PD or } V_s}$

$$= \frac{V_c}{V_s}$$

Since $p_1 V_1^n = p_2 V_4^n$, $\therefore V_1 = V_4 \left(\frac{p_2}{p_1} \right)^{\frac{1}{n}}$

$$= V_c \left(\frac{p_2}{p_1} \right)^{\frac{1}{n}}$$

$$\therefore \eta_{\text{vol}} = 1 + C - \frac{V_c}{V_s} \left(\frac{p_2}{p_1} \right)^{\frac{1}{n}}$$

$$= 1 + C - C \left(\frac{p_2}{p_1} \right)^{\frac{1}{n}} \quad (18.13)$$

Equation (18.13) is plotted in Fig. 18.6. Since (p_2/p_1) is always greater than unity, it is evident that the volumetric efficiency decreases as the clearance increases and as the pressure ratio increases.

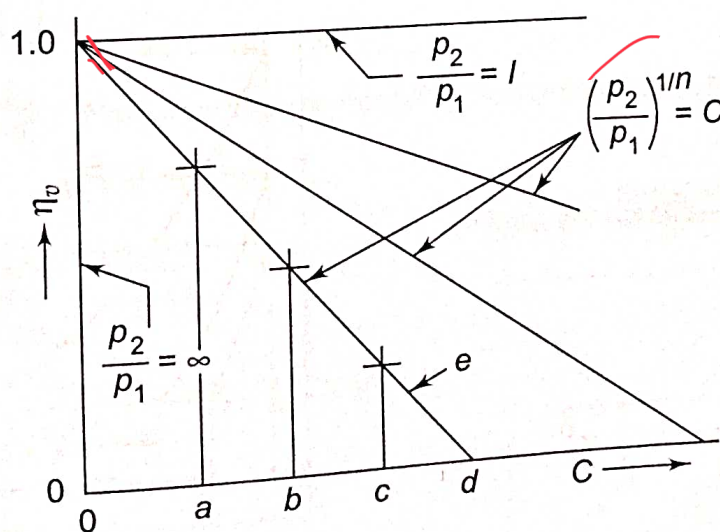


Fig. 18.6 Effect of clearance on volumetric efficiency

In order to get maximum flow capacity, compressors are built with the minimum practical clearance. Sometimes, however, the clearance is deliberately increased

$\left(\dot{m} = \frac{V_s n_v}{v_1}\right)$ as a means of controlling the flow through a compressor driven by a constant speed motor. The compressor cylinder of Fig. 18.7 is fitted with a clearance pocket which can be opened by a valve. Let us suppose that this machine is operating at conditions corresponding to line *e* in Fig. 18.6. If the clearance volume is at minimum value *a* the volumetric efficiency and the flow through the machine are maximum. If the clearance pocket is then opened to increase the clearance to *b*, the volumetric efficiency and the flow are reduced. By increasing the clearance in steps, as indicated by points *c* and *d*, the flow may be reduced in steps to zero. The work per kg of gas compressed is, however, not affected by the clearance volume in an idealised compressor.

For a given pressure ratio, η_{vol} is zero when the maximum clearance is

$$C_{max} = \frac{1}{\left(\frac{p_2}{p_1}\right)^{\frac{1}{n}} - 1} \quad (18.14)$$

18.4.1 Effect of Pressure Ratio on Volumetric Efficiency

It is evident from Fig. 18.6 that as the pressure ratio is increased the volumetric efficiency of a compressor of fixed clearance decreases, eventually becoming zero. This can also be seen in an indicator diagram, Fig. 18.8. As the discharge pressure is increased from p_{2a} to p_{2b} , the volume V_{1a} taken at pressure p_1 decreases to V_{1b} , and η_{vol} decreases. At some pressure p_{2c} the compression line intersects the line of clearance volume and there is no discharge of gas. An attempt to pump to p_{2c} (or any higher pressure) would result in compression and re-expansion of the same gas repeatedly, with no flow in or out.

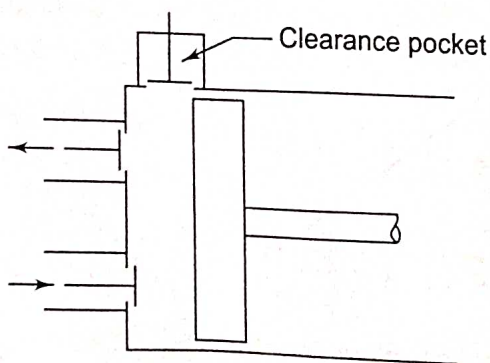


Fig. 18.7 Clearance pocket for capacity control

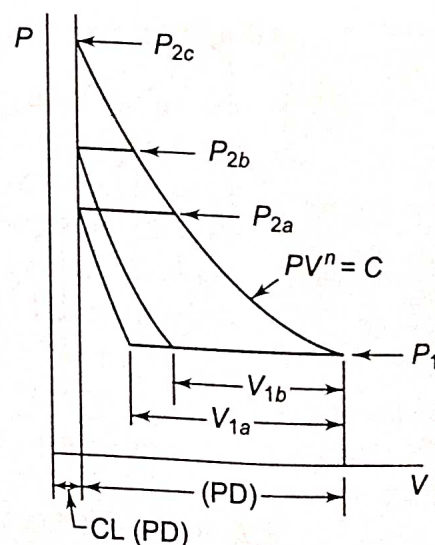


Fig. 18.8 Effect of pressure-ratio on capacity

The maximum pressure ratio attainable with a reciprocating compressor cylinder is then seen to be limited by the clearance, which is given from Eq. 18.13, as

$$\frac{P_{2\max}}{P_1} = \left(1 + \frac{1}{C}\right)^n \quad (18.15)$$

18.5 MULTI-STAGE COMPRESSION

When compressing a gas (or air) to high pressure it is advantageous to do it in stages. The condition for minimum work requires the compression to be isothermal. Since the temperature after compression is given by $T_2 = T_1 (p_2/p_1)^{(n-1)/n}$, the delivery temperature, T_2 , increases with the pressure ratio. Also the volumetric efficiency as given by Eq. (18.13) decreases as the pressure ratio increases, as mentioned earlier.

The volumetric efficiency can be improved by carrying out the compression in two stages. After the first stage of compression (LP) from the state p_1, T_1 to the state p_x, T_x , the fluid is passed into a smaller cylinder (HP) in which the gas is compressed to the required final pressure p_2 at temperature T_2 (Fig. 18.9).

LO 18.5

Explain the details of using multi-stage compression and its preference over single stage

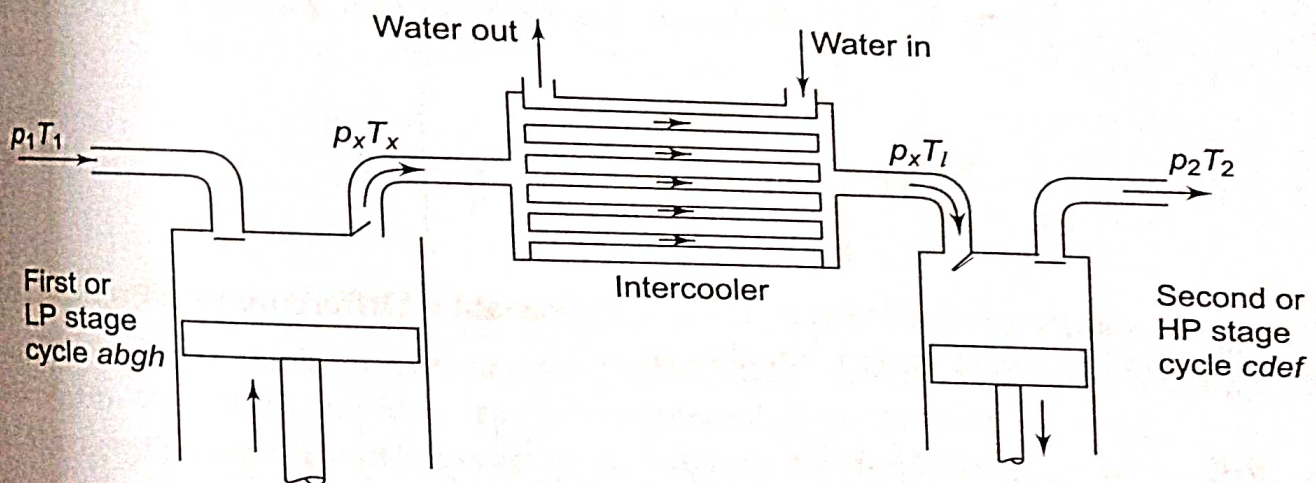


Fig. 18.9 Plan showing intercooling between compressor stages

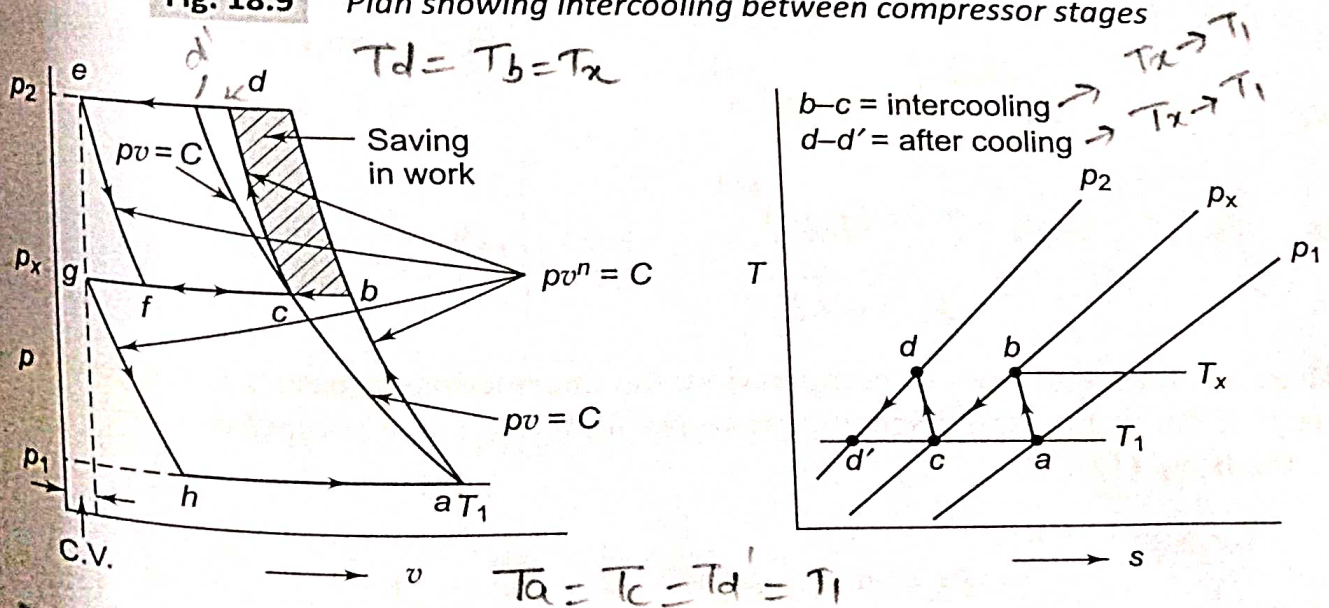


Fig. 18.10 p - V and T - s diagrams for two-stage compression with perfect intercooling, showing the work saved

The gas after being compressed in the LP (low pressure) cylinder ($a-b$) is passed on to an intercooler for getting cooled. After leaving the intercooler, the gas enters the HP (high pressure) cylinder for further compression ($c-d$). Figure 18.10 shows the p - V diagram for two-stage compression. Complete or perfect intercooling ($b-c$) means that the exiting gas from the intercooler at temperature T_x is cooled completely to the original (inlet) temperature T_1 .

Similarly, perfect aftercooling ($d-d'$) makes the gas, leaving the HP compressor, cooled also to the inlet temperature T_1 . This aftercooling reduces the volume of the gas leaving, and thus the size of the receiver becomes smaller. The clearance volume in both the cylinders has here been assumed to be the same.

18.5.1 Ideal Intermediate Pressure

The intermediate pressure p_x (Fig. 18.10) has an optimum value for minimum work of compression. The total work per kg of gas is given by

$$\begin{aligned} W_c &= \frac{n}{n-1} RT_1 \left[\left(\frac{p_x}{p_1} \right)^{\frac{n-1}{n}} - 1 \right] + \frac{n}{n-1} RT_1 \left[\left(\frac{p_2}{p_x} \right)^{\frac{n-1}{n}} - 1 \right] \\ &= \frac{n}{n-1} RT_1 \left[\left(\frac{p_x}{p_1} \right)^{\frac{n-1}{n}} + \left(\frac{p_2}{p_x} \right)^{\frac{n-1}{n}} - 2 \right] \end{aligned} \quad (18.16)$$

Here p_1 , T_1 and p_2 are fixed and p_x is the only variable. Differentiating Eq. (18.16) with respect to p_x and making it equal to zero,

$$\begin{aligned} \frac{dW_c}{dp_x} &= \frac{n}{n-1} RT_1 \left[\frac{n-1}{n} \left(\frac{1}{p_1} \right)^{\frac{n-1}{n}} (p_x)^{-\frac{1}{n}-1} + p_2^{\frac{n-1}{n}} \cdot p_x^{-1+\frac{1}{n}-1} \left(-\frac{n-1}{n} \right) \right] = 0 \\ \therefore p_x^{-\frac{1}{n}+2-\frac{1}{n}} &= (p_2 p_1)^{\frac{n-1}{n}} \\ \text{or } p_x^{2\left(\frac{n-1}{n}\right)} &= (p_1 p_2)^{\frac{n-1}{n}} \\ \therefore p_x &= \sqrt{p_1 p_2} \end{aligned} \quad (18.17)$$

Thus, for minimum work of compression, the intermediate pressure is the geometric mean of the suction and discharge pressures for a two-stage compressor.

From Eq. (18.17)

$$\frac{p_x}{p_1} = \frac{p_2}{p_x} = \left(\frac{p_2}{p_1} \right)^{\frac{1}{2}} \quad (18.18)$$

Pressure ratio in LP stage = Pressure ratio in HP stage

Also,
$$\frac{T_x}{T_1} = \left(\frac{p_x}{p_1}\right)^{\frac{n-1}{n}} \text{ and } \frac{T_2}{T_1} = \left(\frac{p_2}{p_x}\right)^{\frac{n-1}{n}}$$

$$\therefore T_2 = T_x \quad (18.19)$$

From Fig. 18.10, it is seen that work required in LP compressor = Work required in HP compressor.

Thus, the intermediate pressure that produces minimum work will also result in equal pressure ratios in the two stages of compression, equal discharge temperatures and equal work for the two stages.

For two stage-compression, the minimum work, using Eq. (18.16) becomes

$$W_c = \frac{2nRT_1}{n-1} \left[\left(\frac{p_x}{p_1}\right)^{\frac{n-1}{n}} - 1 \right] \quad (18.20)$$

where p_x/p_1 is the pressure ratio in each stage. In terms of overall pressure ratio, it becomes

$$W_c = \frac{2nRT_1}{n-1} \left[\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{2n}} - 1 \right] \quad (18.21)$$

Heat rejected in the intercooler, $Q_{bc} = c_p (T_x - T_1)$ kJ/kg

Let us now consider compression efficiencies and imperfect intercooling. As shown in Fig. 18.11, an ideal gas is compressed from the initial state p_1, T_1 to p_x . It is then cooled at constant pressure to T_y and then compressed from p_x, T_y to p_2 . Given p_1, T_1, T_y and p_2 , it is desired to find the value of p_x which gives minimum work. Let the adiabatic compression efficiencies of the two stages be, respectively, η_{c1} and η_{c2} ; then the work of compression is

$$\begin{aligned} W_c &= W_1 + W_2 \\ &= \frac{1}{\eta_{c1}} \frac{\gamma}{\gamma-1} RT_1 \left[\left(\frac{p_x}{p_1}\right)^{\frac{\gamma-1}{\gamma}} - 1 \right] + \frac{1}{\eta_{c2}} \frac{\gamma}{\gamma-1} RT_y \left[\left(\frac{p_2}{p_y}\right)^{\frac{\gamma-1}{\gamma}} - 1 \right] \end{aligned}$$

But

$$\left(\frac{p_x}{p_1}\right)^{\frac{\gamma-1}{\gamma}} = \frac{T'_x}{T_1} \text{ and } \left(\frac{p_2}{p_y}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{p_2}{p_x}\right)^{\frac{\gamma-1}{\gamma}} = \frac{T'_2}{T_x}$$

$\therefore$

$$W_c = \frac{\gamma R}{\gamma-1} \left[\frac{T_1}{\eta_{c1}} \left(\frac{T'_x}{T_1} - 1 \right) + \frac{T_y}{\eta_{c2}} \left(\frac{T'_2}{T_x} - 1 \right) \right]$$

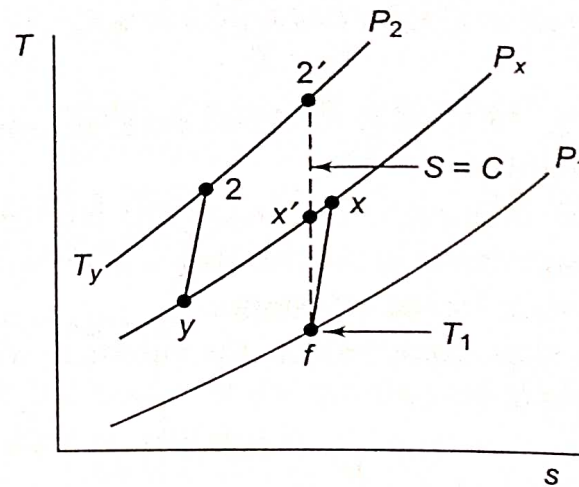
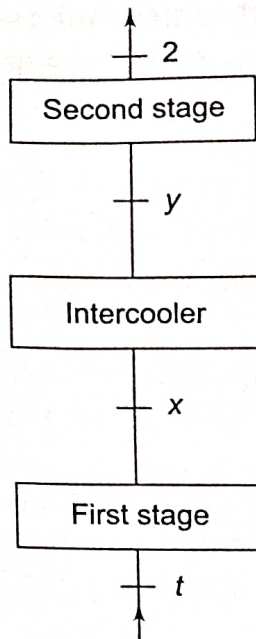


Fig. 18.11 *T-s plot of two-stage compression process*

Taking the derivative with respect to T'_x and setting it equal to zero (noting that T_1 , T'_2 and T_y are constant),

$$\frac{dW_c}{dT'_x} = 0 = \frac{\gamma R}{\gamma - 1} \left[\frac{1}{\eta_{C1}} + \frac{T_y T'_2}{\eta_{C2}} \left(-\frac{1}{(T'_x)^2} \right) \right]$$

Then,

$$(T'_x)^2 = \frac{\eta_{C1}}{\eta_{C2}} T_y T'_2$$

and

$$\frac{T'_x}{T_1} = \sqrt{\frac{\eta_{C1}}{\eta_{C2}} \cdot \frac{T_y}{T_1} \cdot \frac{T'_2}{T_1}}$$

for minimum work. Now

$$\frac{T'_x}{T_1} = \left(\frac{p_x}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \quad \text{and} \quad \frac{T'_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}}$$

Therefore, for minimum work in two-stage compression of an ideal gas with intercooling to a fixed temperature T_y ,

$$\frac{p_x}{p_1} = \sqrt{\left(\frac{\eta_{C1}}{\eta_{C2}} \frac{T_y}{T_1} \right)^{\frac{\gamma}{\gamma-1}} \cdot \frac{p_2}{p_1}} \quad (18.22)$$

For the special case of $T_y = T_1$ and $\eta_{C1} = \eta_{C2}$, which is often taken as a standard of comparison, the requirement for minimum work is as obtained earlier in Eq. (18.18).

$$\frac{p_x}{p_1} = \sqrt{\frac{p_2}{p_1}} \quad (18.23)$$

Also for this special case, the condition of minimum work is the condition of equal work in the two stages.

When, three stages of equal efficiencies are used, with intercooling to the initial temperature at two points as shown in Fig. 18.12, the condition of minimum work, and of equal division of work among stages is

$$\frac{p_{x1}}{p_1} = \frac{p_{x2}}{p_{x1}} = \frac{p_2}{p_{x2}} = \left(\frac{p_2}{p_1} \right)^{1/3} \quad (18.24)$$

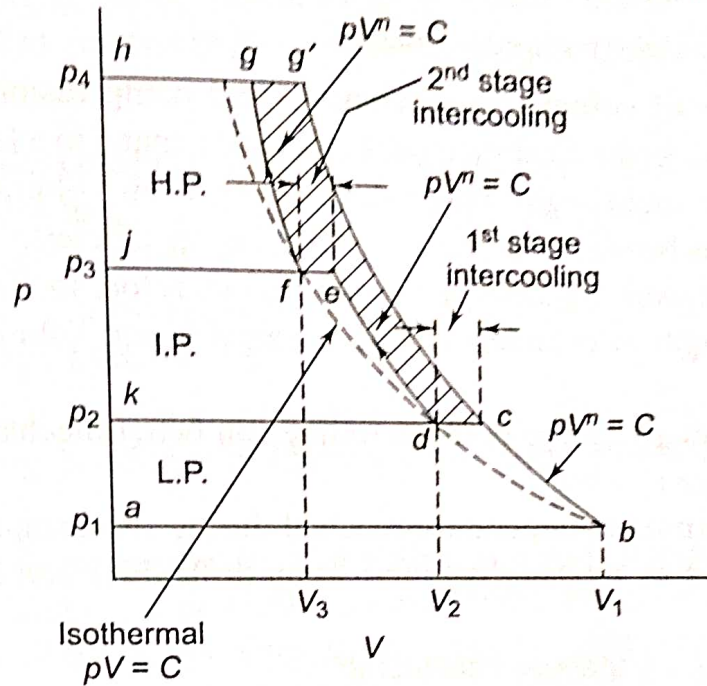


Fig. 18.12 Three-stage compression with perfect intercooling

Thus for three-stage compressor the optimum pressure ratio per stage can be written as

$$\frac{p_x}{p_1} = \left(\frac{p_d}{p_s} \right)^{\frac{1}{3}}$$

where p_d and p_s are the discharge and suction pressure, respectively.

For N-stage compressor, the optimum pressure ratio per stage is

$$\frac{p_x}{p_1} = \left(\frac{p_d}{p_s} \right)^{\frac{1}{N}} \quad (18.25)$$

The minimum work of compression for N-stages is then

$$W_c = \frac{NnRT_1}{n-1} \left[\left(\frac{p_d}{p_s} \right)^{\frac{n-1}{Nn}} - 1 \right] \quad (18.26)$$

In the case of gas compression, the desirable idealised process is, however, a reversible isothermal process, and the isothermal efficiency is given by

$$\eta_t = \frac{W_t}{W_c} = \frac{p_1 v_1 \ln \frac{p_2}{p_1}}{W_c} \quad (18.27)$$

The indicator diagram for a three-stage compressor with perfect intercooling is shown in Fig. 18.12. The air delivered from the first stage at c is cooled to point d on the curve $pV = C$ (shown dotted) before admission to the second stage cylinder. Again the air is cooled from e to f after compression in the second stage and before admission to the third stage cylinder. The shaded area represents the saving of work resulting from such intercooling since without intercooling the compression curve would have followed the path bca' .

Advantages of multi-stage compression are:

1. Improved overall volumetric efficiency. If all compression was done in one cylinder the gas in the clearance volume would expand to a large volume before the new intake could begin. This results in a very low volumetric efficiency. By cooling the gas between the stages a much higher efficiency can be obtained.
2. A reduction in work required per stroke, and therefore the total driving power.
3. Size and strength of cylinders can be adjusted to suit volume and pressure of gas.
4. Multi-cylinders give more uniform torque and better mechanical balance thus needing smaller flywheel.
5. Since the maximum temperature reached during the compression process is greatly reduced by intercooling lubrication difficulties and explosion hazards are lessened.
6. Leakage losses are reduced considerably.

Practise appears to indicate that the economical value of pressure ratio per stage is in the range of 3 to 5. For compression of atmospheric air, single-stage machines are often used up to 550 kPa discharge pressure, two-stage from 350 kPa to 2.1 MPa, three stage from 2.1 MPa to 7.0 MPa and four or more stages for higher pressures.